DANG TRAN NAM

Software Engineer · Ho Chi Minh City, Vietnam

nam.dt161@gmail.com · 0981405314 · GitHub · LinkedIn

Detail-oriented Software Engineer with a strong background in backend systems, database architecture, and full-stack SaaS performance. I build reliable, high-efficiency software with clean code, data-driven latency optimization, and robust distributed resilience.

Experience

An Binh Hospital

Software Engineer Intern — IT Asset Management (ITAM): Distributed Telemetry System

- **Distributed Telemetry Agent:** Designed and developed a cross-platform Go agent harvesting real-time CPU, RAM, and Disk metrics across 100+ edge devices for centralized observability.
- **Resilience & Reliability:** Implemented Circuit Breaker pattern with Exponential Backoff and Jitter to absorb thundering-herd reconnect spikes, maintaining 99.9% uptime.
- **Performance Optimization:** Reduced agent memory footprint by over 60% (under 15MB) using log rotation and memory-efficient data structures.
- **Security:** Secured endpoint telemetry with Bearer Token auth and SHA-256 HMAC integrity verification.

Key Projects

EzSplit — Full-Stack AI Linguistic SaaS

Problem: Memorizing vocabulary without understanding etymological structure leads to fast forgetting.

Solution: Built full-stack SaaS with morpheme segmentation, Bézier etymology trees, SM-2 Spaced Repetition, native English Web Speech API audio, and automated PayOS VietQR + PayPal checkout.

Result: Deployed live SaaS with 51 static/dynamic routes, sub-second latency, and bilingual localization.

Live Demo: ezsplit.vercel.app

minPOS — Offline-First Micro-Merchant POS

Problem: Small retail/F&B merchants need an ultra-fast, affordable POS system that works reliably without internet.

Solution: Embedded Go backend with SQLite local storage, background transactional sync, thermal receipt printing, and hardware-bound license key generator.

Result: Sub-50ms transaction speeds, offline-first reliability, zero data loss.

GitHub Repository

High-Concurrency Inventory Engine

Problem: Flash-sale peak traffic caused concurrency race conditions and inventory overselling.

Solution: Optimistic Locking with version columns, ACID transactions, and composite indexing.

Result: 50% query time reduction, 0 overselling incidents under high concurrent load.

GitHub Repository

build-a-pc — AI PC Compatibility Engine

Problem: Choosing matching computer components requires complex validation across specs and bottlenecks.

Solution: Part compatibility engine with Gemini API goal-based recommendations and dynamic SEO pages.

Result: 8 shipped iterations, active affiliate links, sub-second hardware matching.

Live Demo: build-a-pc.vercel.app

Technical Skills

Languages: Go (Golang), TypeScript, JavaScript, Java, C/C++, SQL

Frameworks & Backend: Next.js, React, Node.js, Express, Prisma ORM, Tailwind CSS

Databases: PostgreSQL, SQLite, Redis

DevOps & Tools: Linux, Docker, GCP, Git, CI/CD, PM2, Circuit Breaker, High-Concurrency Architectures

Education

Saigon University (SGU) – Software Engineering (2021 – 2026)
Engineer Degree

Certifications

- TOEIC (760 R/L)
- AWS Cloud (In Progress)